ModelIt!

Student Activity Pack

The Invisible War: How Your Immune System Fights Back

Your Body's Army Against Invaders

Grade 07L2 | NGSS MS-LS1-3, MS-LS1-8 | Systems and System Models

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think happens inside your body when you get sick?

2 Why do you think some people get sicker than others from the same illness?

3 Draw what you think a white blood cell does when a virus enters your body.

4 How do you think vaccines help prevent disease?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Pathogen Load | Vaccine Status | White Blood Cell Response | Antibody Production | Infection Severity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Unvaccinated + High Pathogen

High pathogen load, no prior vaccination

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Vaccinated + High Pathogen

High pathogen load, immune system pre-trained

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Low Pathogen + No Vaccine

Small pathogen exposure, no vaccine

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Memory Cell Formation — Long-lived immune cells that remember specific pathogens and activate faster during re-infection

Fever Response — The body raising its temperature to slow pathogen reproduction and speed up immune cell activity

Inflammatory Response — Increased blood flow and swelling at the infection site that brings more immune cells to the area

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does your body know the difference between you and an invader, and how does it fight back? Use evidence from your simulation to support your answers.

1. **Why does it take several days to recover from a new illness but your body fights it faster the second time?**

2. **How does your model show that vaccines don't prevent exposure but do prevent severe illness?**

3. **What would happen in your model if white blood cells couldn't communicate with antibody-producing cells?**

4. **Why is herd immunity important for people who can't be vaccinated?**

5. **How is the immune system an example of interacting subsystems within the body?**

8 STEM Challenge: Design a Public Health Campaign

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YOUR ENGINEERING MISSION

Create an evidence-based public health campaign explaining how the immune system works and why vaccination matters, using data from your simulation model.

THE SCENARIO

A local health department needs help explaining to the community how vaccines work. Your team has been asked to create educational materials that use real scientific models to show the difference between vaccinated and unvaccinated immune responses.

Engineering Design Thinking

1. WHAT FORMAT WILL YOUR CAMPAIGN USE (POSTER, VIDEO, INFOGRAPHIC, PRESENTATION)?
2. HOW WILL YOU REPRESENT THE INVISIBLE PROCESSES OF THE IMMUNE SYSTEM VISUALLY?
3. WHAT MODEL DATA WILL YOU INCLUDE AS EVIDENCE FOR VACCINE EFFECTIVENESS?
4. HOW WILL YOU MAKE COMPLEX IMMUNOLOGY ACCESSIBLE TO A GENERAL AUDIENCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Immunologists study the immune system and develop treatments for diseases, allergies, and autoimmune disorders. They work at research hospitals, pharmaceutical companies, and the CDC, earning $75,000-$150,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 A vaccinated person and an unvaccinated person are both exposed to the same high dose of a virus. The vaccinated person recovers in two days with mild symptoms, while the unvaccinated person is sick for ten days. Based on the immune system model, which explanation best accounts for this difference?

- **A.** The vaccinated person has a stronger physical barrier that prevents pathogens from entering the body
- **B.** The vaccinated person's memory cells enable rapid antibody production, neutralizing the pathogen before it can spread widely
- **C.** The vaccine destroyed the virus before it could enter the vaccinated person's body
- **D.** The unvaccinated person has fewer white blood cells than the vaccinated person

2 In the immune system model, when Pathogen Load is set to high and Vaccine Status is off, Infection Severity rises rapidly before White Blood Cell Response can contain the invasion. Which relationship in the model best explains this outcome?

- **A.** Pathogen Load has a negative relationship with Infection Severity
- **B.** White Blood Cell Response has a positive relationship with Infection Severity
- **C.** Pathogen Load has a positive relationship with Infection Severity, and antibody production is delayed without prior vaccination
- **D.** Vaccine Status has a negative relationship with White Blood Cell Response

3 A student claims that white blood cells and antibodies perform the same function in the immune system. Using the model, which response best corrects this misconception?

- **A.** White blood cells provide a fast, general response against any invader, while antibodies are custom-built proteins that specifically target one pathogen type
- **B.** White blood cells only work during vaccinated responses, while antibodies work during unvaccinated responses
- **C.** Antibodies arrive first and white blood cells arrive later to clean up remaining pathogens
- **D.** White blood cells and antibodies are the same thing but are measured differently in the model

4 In the model, setting Pathogen Load to low with Vaccine Status off still results in low Infection Severity. Which explanation best accounts for why the body can handle a small invasion without prior vaccination?

- **A.** Low pathogen loads do not trigger any immune response
- **B.** The non-specific white blood cell response is sufficient to contain a small number of pathogens before they can multiply significantly
- **C.** The body only uses antibodies against small pathogen loads
- **D.** Low pathogen loads cannot cause any disease regardless of immune system function

5 **A community has 90% vaccination rates for a particular virus. A student who cannot be vaccinated due to a medical condition is protected from the disease. Which concept from the model best explains this protection?**

A. The unvaccinated student develops antibodies by breathing the same air as vaccinated students

B. When most people in a community are immune, the pathogen cannot spread easily enough to reach vulnerable individuals, which is known as herd immunity

C. Vaccines release protective chemicals into the environment that shield nearby unvaccinated people

D. The unvaccinated student's white blood cells are strengthened by proximity to vaccinated individuals